🛠 Phase 1: Download & Install the Tools

You need to install these programs on your computer first. **Install them in this exact order.**

1. GitHub Desktop (To sync our code)

- You should already have this! If not, download it from desktop.github.com and log in with your GitHub account.

2. Python (For our Backend API)

- Go to python.org and download the latest version.
- 🚨 **CRUCIAL STEP:** When you open the installer, look at the very bottom of the window and **check the box that says "Add Python.exe to PATH"** before you click Install. If you miss this, the backend won't work!

3. XAMPP (For our MySQL Database)

- Go to apachefriends.org and download XAMPP for Windows.
- Install it and keep pressing **Next** (leave all default settings).

4. Visual Studio Code (VS Code)

- Go to code.visualstudio.com and download it. This is the text editor where we will actually write and view our code.

5. Android Studio (For the Virtual Phone)

- Go to developer.android.com/studio and download Android Studio.
- Run the installer. Make sure **Android Virtual Device** is checked during installation.
- Open Android Studio once, let it download its initial updates, and click **Finish**. We just need it installed in the background so we can use its virtual phones.

6. Flutter (For our Frontend App)

- Go to the Flutter Windows install page and download the Flutter SDK zip file.
- Extract the zip folder and place the `flutter` folder directly in your `C:\` drive (so it looks like `C:\flutter`).
- **Add Flutter to your computer's PATH:**
    1. Click your Windows Start button, type `Environment Variables`, and hit Enter.
    2. Click the **Environment Variables...** button at the bottom.
    3. Under "User variables" (the top box), find the one named **Path**, click it, and click **Edit**.

4. Click **New** and type exactly this: C:\flutter\bin
5. Click OK on all windows to save and close them.

📥 Phase 2: Get the Code (Using GitHub Desktop)

Now let's download the project to your computer.

1. Open **GitHub Desktop**.
2. Click **File** at the top left, then click **Clone repository...**
3. Click the **URL** tab.
4. Paste our project's GitHub link into the top box.
5. Choose where you want to save the folder on your computer (e.g., your Documents folder).
6. Click **Clone**.
7. Once it's done, open **VS Code**, click **File > Open Folder...**, and select the project folder you just downloaded.

🗄 Phase 3: Set Up the Database

Our app needs a place to store student accounts and votes. We need to import the team's database structure into your local XAMPP.

1. Open the **XAMPP Control Panel** from your Windows Start menu.
2. Click the **Start** button next to **Apache**.
3. Click the **Start** button next to **MySQL**. *(Both words should turn green with a PID number next to them).*
4. Click the **Admin** button right next to MySQL. This will open a webpage in your browser called phpMyAdmin.
5. On the left side of the screen, click **New**.
6. Type exactly **emobile_voting** in the Database name box and click **Create**.
7. Click on your new emobile_voting database on the left side menu to select it.
8. At the top of the screen, click the **Import** tab.
9. Click **Choose File** and locate the .sql file inside our downloaded project folder.
10. Scroll all the way down and click **Import** (or **Go**). You should see a green success message!

⚙ Phase 4: Run the Python Backend

The backend is the brain of our app. Let's turn it on.

1. In VS Code, click **Terminal** at the very top menu bar, then click **New Terminal**.
2. A small window will pop up at the bottom. Type this to move into the backend folder and press Enter:

   ```
   cd backend
   ```

3. Install the required Python libraries (`fastapi`, `uvicorn`, `sqlalchemy`, `pymysql`, `passlib[bcrypt]`, `pyjwt`, `pydantic`) by pasting this command and pressing Enter:

   ```
   pip install -r requirements.txt
   ```

4. Turn on the server by typing:

   ```
   uvicorn main:app --reload
   ```

5. If it says "Application startup complete", your backend is successfully running! **Leave this terminal open.**

📱 Phase 5: Run the Flutter App

Now let's launch the actual visual app.

1. In VS Code, open a **Second Terminal** (Click the + icon in the top right of the terminal window so you don't close your Python server).
2. Move into the frontend folder:

   ```
   cd frontend
   ```

3. Download the Flutter packages:

   ```
   flutter pub get
   ```

📱 **To run the app as a Website (Chrome):** Type this into the terminal and press Enter:

```
flutter run -d chrome
```

📱 **To run the app on an Android Phone:**

1. Open the Command Palette in VS Code by pressing `Ctrl + Shift + P`.
2. Type `Flutter: Launch Emulator` and press Enter.
3. Select your Pixel emulator from the list and wait 1-2 minutes for the virtual phone to appear and turn on.
4. Once the phone is on the home screen, type this into your terminal:

```
flutter run
```

🔄 Daily Routine: Saving and Syncing Code

Because we are working as a team, you must sync your code using **GitHub Desktop**.

Before you start working every day:

1. Open GitHub Desktop.
2. Click **Fetch origin** at the top right. If it changes to **Pull origin**, click it! This downloads the team's latest updates to your computer.

When you finish working and want to share your code:

1. Open GitHub Desktop. You will see a list of files you changed on the left.
2. In the bottom left box (Summary), type a short message about what you did (e.g., "Designed the login button").
3. Click the blue **Commit to main** button.
4. Finally, click **Push origin** at the top right to upload your changes to the internet for the rest of the team!